Task 3. Celebrity Problem

Available marks: 21

A *celebrity* is someone whom everybody knows, but who doesn't know anybody (because they don't need to).

Consider a matrix that describes the relationships between *N* people. The cell at row *i* and column *j*, counting from the top left corner, is either a **1** if person *i* knows person *j*, or a **0** if person *i* does not know person *j*. Values on the major diagonal are irrelevant as philosphers have been arguing for centuries about whether one can know oneself .

Example Matrix

	1	2	3	4	5	6
1		0	0	1	0	1
2	0		0	1	1	0
3	1	0		1	0	0
4	0	0	0		0	0
5	0	1	0	1		1
6	0	0	1	1	1	

The matrix shows that person 3 knows person 1 (because row 3 col 1 is a **1**), but person 1 does not know person 3. Person 4 is the celebrity.

Your Task

Given a matrix with up to 50 rows and columns, determine whether it represents a group of people that includes a celebrity, and if so, what the celebrity's person number is (persons count from 1 to *N*). This is not very difficult, but to get full marks your program has to be smart about the search. While a brute-force solution inspects about half the cells in the matrix (or around $N^2/2$ cells on average), for full marks your solution must inspect a maximum of $3N–3$ cells, no matter where the celebrity is or if there is one at all.

Data Format

The first line of input contains the matrix order, *N*, which lies between 2 and 50. Then follows *N* lines, each with *N* elements each separated by a space. An element is either 1 or 0, or – for the diagonal element. For example, the matrix above is available in file **task3example.txt**, which contains:

```
6
- 0 0 1 0 1
0 - 0 1 1 0
1 0 - 1 0 0
0 0 0 - 0 0
0 1 0 1 - 1
0 0 1 1 1 -
```

The program must report the number of cells inspected as well as the result. You may be required to explain your algorithm and how this number was calculated, in the event that you claim to have produced a linear-time solution.

Assessment

When you believe you have completed this task the judges will ask you to run the program with some of the data files in the task directory, at their discretion.

- **11 marks** are awarded if you can correctly identify the celebrity where it exists, regardless of the algorithm used.

- An additional **10 marks** are awarded if your program, to the satisfaction of the judges, also works in linear time in the worst case. Your program must produce a correct solution in all cases before this mark is available.

- If you are stuck on this task, you may purchase a hint that will *probably* allow you to produce a linear solution, though you'll still have to do some thinking. This hint costs **5 marks**, deducted from your total.